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Ride Sharing System

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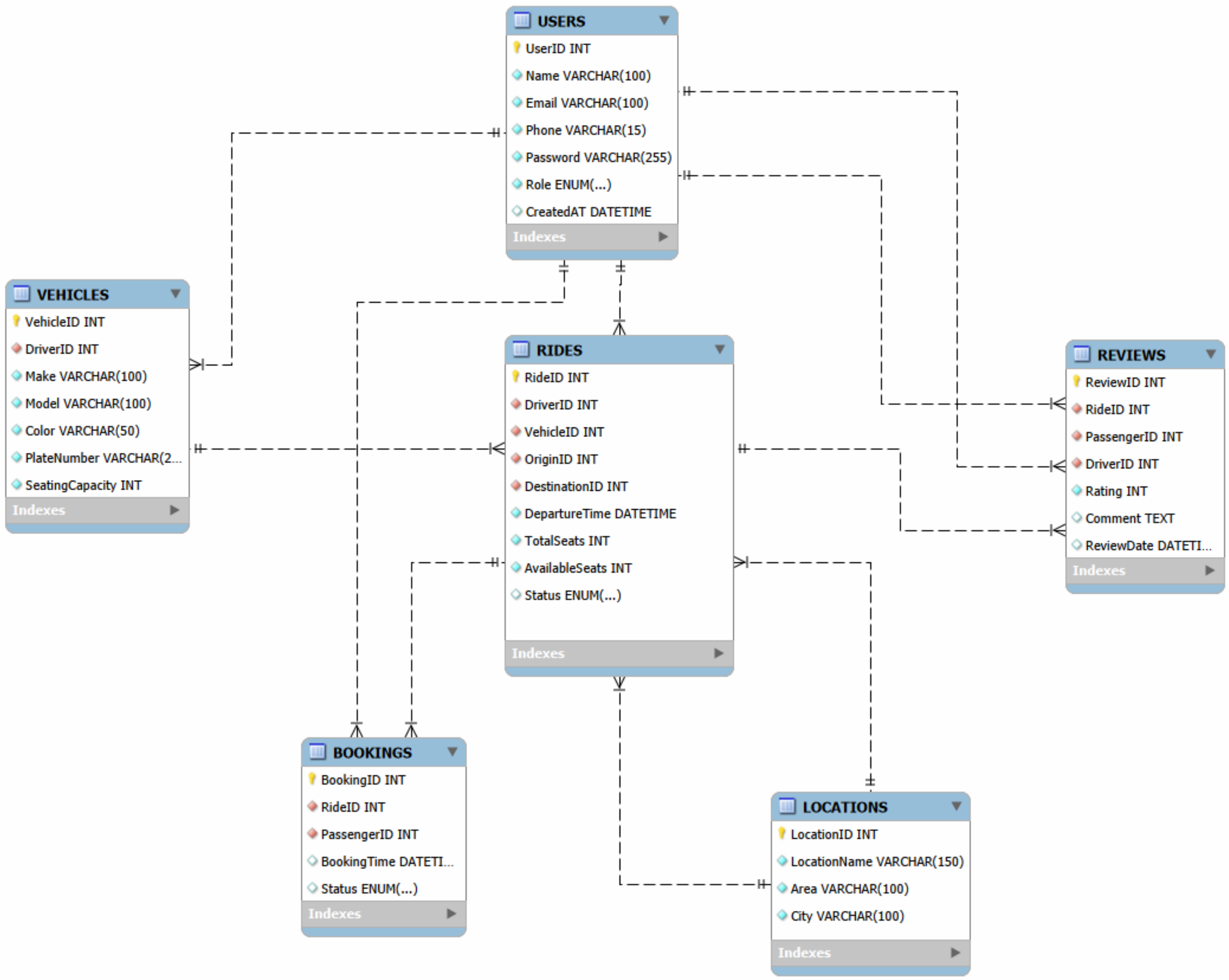
Project Milestones

Milestone	Version	Date	Remarks
Milestone 1 – ERD & Relational Schema	V1.0	26/04/2026	

1. Entity-Relationship Diagram (ERD)



Entity-Relationship Diagram – Ride Sharing System

2. Relational Database Schema

This section presents the relational schema derived from the Entity-Relationship Diagram (ERD). Each entity is transformed into a table with clearly defined primary keys (PK) and foreign keys (FK) to maintain relationships and data integrity.

2.1 USERS

Description: Stores all system users, including both drivers and passengers.

```
USERS( UserID PK, Name, Email, Phone, Password, Role, CreatedAt )
```

Field	Type	Null	Key	Default	Extra
UserID	int	NO	PRI	NULL	auto_increment
Name	varchar(100)	NO		NULL	
Email	varchar(100)	NO	UNI	NULL	
Phone	varchar(15)	NO		NULL	
Password	varchar(255)	NO		NULL	
Role	enum('Driver','Passenger')	NO		NULL	
CreatedAt	datetime	NO		NULL	

USERS Table

2.2 VEHICLES

Description: Contains vehicle details associated with drivers.

```
VEHICLES( VehicleID PK, DriverID FK -> USERS(UserID), Make, Model, Color,
PlateNumber, SeatingCapacity )
```

Field	Type	Null	Key	Default	Extra
VehicleID	int	NO	PRI	NULL	auto_increment
DriverID	int	NO	MUL	NULL	
Make	varchar(100)	NO		NULL	
Model	varchar(100)	NO		NULL	
Color	varchar(50)	NO		NULL	
PlateNumber	varchar(20)	NO		NULL	
SeatingCapacity	int	NO		NULL	

VEHICLES Table

2.3 LOCATIONS

Description: Stores pickup and destination location details.

```
LOCATIONS( LocationID PK, LocationName, Area, City )
```

Field	Type	Null	Key	Default	Extra
LocationID	int	NO	PRI	NULL	auto_increment
LocationName	varchar(150)	NO		NULL	
Area	varchar(100)	NO		NULL	
City	varchar(100)	NO		NULL	

LOCATIONS Table

2.4 RIDES

Description: Represents rides created by drivers, including route, timing, and seat availability.

```
RIDES( RideID PK, DriverID FK -> USERS(UserID), VehicleID FK ->
VEHICLES(VehicleID), OriginID FK -> LOCATIONS(LocationID), DestinationID FK -
> LOCATIONS(LocationID),DepartureTime, TotalSeats, AvailableSeats, Status )
```

Field	Type	Null	Key	Default	Extra
RideID	int	NO	PRI	NULL	auto_increment
DriverID	int	NO	MUL	NULL	
VehicleID	int	NO	MUL	NULL	
OriginID	int	NO	MUL	NULL	
DestinationID	int	NO	MUL	NULL	
RideTime	date	NO		NULL	
DepartureTime	time	NO		NULL	
TotalSeats	int	NO		NULL	
AvailableSeats	int	NO		NULL	
Status	enum('Scheduled','Ongoing','Completed')	NO		NULL	

RIDES Table

2.5 BOOKINGS

Description: Stores ride reservations made by passengers.

```
BOOKINGS( BookingID PK, RideID FK -> RIDES(RideID), PassengerID FK ->
USERS(UserID), BookingTime, Status )
```

Field	Type	Null	Key	Default	Extra
BookingID	int	NO	PRI	NULL	auto_increment
RideID	int	NO	MUL	NULL	
PassengerID	int	NO	MUL	NULL	
BookingTime	datetime	NO		NULL	
Status	enum('Pending','Confirmed','Cancelled')	NO		NULL	

BOOKINGS Table

2.6 REVIEWS

Description: Stores feedback and ratings given by passengers to drivers after rides.

```
REVIEWS( ReviewID PK, RideID FK -> RIDES(RideID), PassengerID FK ->
USERS(UserID), DriverID FK -> USERS(UserID), Rating, Comment, ReviewDate )
```

Field	Type	Null	Key	Default	Extra
ReviewID	int	NO	PRI	NULL	auto_increment
RideID	int	NO	MUL	NULL	
PassengerID	int	NO	MUL	NULL	
DriverID	int	NO	MUL	NULL	
Rating	int	NO		NULL	
Comment	text	YES		NULL	
ReviewDate	datetime	NO		NULL	

REVIEWS Table